

MICHAEL BARON

Department of Mathematics and Statistics

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Highlights

- Fellow of The American Statistical Association (2013)
- Elected Member of the International Statistical Institute (2024)
- Abraham Wald Best Paper Award in Sequential Analysis (2007)
- Regents' Outstanding Teaching Award (2014)
- Sustained NSF and other external research funding as PI and sole PI
- Author of *Probability and Statistics for Computer Scientists*, now in its third edition and translated into Chinese
- 12 supervised Ph.D. students, pursuing academic and research careers in academia and medicine
- Associate Editor of *Sequential Analysis* and *Applied Stochastic Models in Business and Industry*
- Organizer and Chair of major international conferences in sequential methodology and related areas
- More than 100 invited presentations, including a plenary lecture at the 8th International Workshop in Sequential Methodologies
- David Carroll Professor in Mathematics and Statistics at American University; former Department Chair

Educational History

- **Ph.D. in Statistics** (1995). The University of Maryland, Baltimore, MD.
Dissertation: Confidence Estimation in the Change-Point Problem.
Advisor: Professor A. L. Rukhin (Mathematical Statistics)
- **M.S. in Mathematics** (1992). State University of St. Petersburg, St. Petersburg, Russia.
Dissertation: On the First Passage Time for Queuing Processes.
Advisor: Professor I. A. Ibragimov (Probability Theory)

Professional Experience

- **David Carroll Professor in Mathematics and Statistics** (2014 – Present), **Department Chair** (2022 – 2024); Department of Mathematics and Statistics, American University, Washington DC, 20016
- **Professor** (2006 – 2016), **Associate Professor** (2001 – 2006), **Assistant Professor** (1995 – 2001); Department of Mathematical Sciences, The University of Texas at Dallas, Richardson, Texas, 75080
- **Academic Visitor** (2003 - 2004); Department of Mathematical Sciences, IBM T. J. Watson Research Center, Yorktown Heights, New York, 10598
- **Instructor** (1995), Department of Mathematics and Statistics, University of Maryland, Baltimore County, Baltimore, Maryland, 21250

Research Areas

- Sequential analysis and optimal sequential design
- Sequential and retrospective change-point detection, including multichannel and transient changes
- Sequential multiple testing and simultaneous inference
- Bayesian decision theory and optimal stopping
- Applications of statistical methodology to epidemiology and clinical trials, statistical quality control and semiconductor manufacturing, actuarial science, cybersecurity, energy finance, and other streaming-data problems

Professional Recognition and Honors

- Fellow of the American Statistical Association (2013)
“For outstanding contributions to the theory and applications of sequential analysis and change-point problems, excellence in teaching and mentoring of doctoral students, dissemination of statistical knowledge via book authorship, and promotion of statistical research in diverse areas of industry.”
- Elected Member of the International Statistical Institute (2024)
“In recognition to significant contributions to the statistical profession.”
- Regents’ Outstanding Teaching Award (2014)
- Abraham Wald Prize for the best paper in the Journal of Sequential Analysis (2007)
- Elected Academic Advisor of Asia Pacific Academy for Multidisciplinary Research and Study (2019)
- Guest Editor of Applied Stochastic Models in Business and Industry (2020)
- Associate Editor of two professional journals
- Invited Technometrics paper of 2001
- Plenary speaker at the 8th International Workshop in Sequential Methodologies
- Over 100 invited presentations

Professional Memberships

- American Statistical Association (lifetime member, Fellow since 2013)
- International Statistical Institute (Elected Member of ISI since 2024)
- Institute of Mathematical Statistics (lifetime member)
- International Society for Bayesian Analysis (lifetime member)

Achievements in Original Investigation and Research

Grants awarded

1. Research grants as PI/Co-PI

- Project: Lifelong Streaming Anomaly Detection (Co-PI)
Sponsor: Defense Advanced Research Projects Agency, US Department of Defense
Amount: \$209,640

Duration: 07/01/2019 – 06/30/2020

- Project: ATD: Statistical Detection of New Patterns and Potential Threats in Geospatial Sequences of Social and Political Events (PI)
Sponsor: National Science Foundation, Mathematical Sciences
Amount: \$200,000
Duration: 09/01/2017 – 08/31/2021
- Project: ATD: Efficient online detection based on multiple sensors, with applications to cybersecurity and discovery of biological threats (sole PI)
Sponsor: National Science Foundation, Mathematical Sciences
Amount: \$273,530
Duration: 12/01/2014 – 09/30/2017
- Project: Statistical methods for gaining precision in credibility estimation (PI)
Sponsor: Actuarial Foundation, Individual Grants Competition
Amount: \$21,461
Duration: 06/01/2014 – 05/31/2016
- Project: ATD: Efficient online detection based on multiple sensors, with applications to cybersecurity and discovery of biological threats (PI)
Sponsor: National Science Foundation, Mathematical Sciences
Amount: \$252,060
Duration: 10/01/2013 – 04/30/2015
- Project: Sequential testing of multiple hypotheses, simultaneous confidence intervals, and multichannel change-point detection (sole PI)
Sponsor: National Science Foundation, Division of Mathematical Sciences
Amount: \$200,000
Duration: 06/01/2010 -- 05/31/2013, extension until 05/31/2014
- Project: Fast detection of epidemic trends and change-point analysis of Markov processes (sole PI).
Sponsor: National Security Agency, Mathematical Sciences
Amount: \$45,459
Duration: 02/11/2011 – 02/10/2013
- Project: Sequential analysis of actuarial risks and credibility (sole PI)
Sponsor: Actuarial Foundation, Individual Grants Competition
Amount: \$10,000
Duration: 06/01/2010 -- 05/31/2012
- Project: Optimal sequential planning (sole PI)
Sponsor: National Security Agency, Mathematical Sciences
Amount: \$43,847

Duration: 02/01/2007 -- 01/31/2009

- Project: Identifying change points and patterns in development, learning and problem solving processes (co-PI)
Sponsor: National Science Foundation
Program: Methodology, Measurement, and Statistics (MMS)
Amount: \$68,386
Duration: 05/01/1999 -- 04/30/2001
- Project: Pattern recognition and signature analysis in semiconductor risk assessment (sole PI)
Sponsor: Texas Instruments, Inc.
Amount: \$20,137
Duration: 08/01/1997 -- 07/31/1998

2. Collaborative grants as Investigator/Senior Personnel/Subcontractor

- Project: Confirmation of the Low Glutamate Diet as a Treatment for Gulf War Illness (Investigator; Lead PI: Kathleen Holton, American University)
Sponsor: U. S. Department of Defense, Gulf War Illness Research Program
Amount: \$6,370,661
Duration: 09/01/2022 – 08/31/2025
- Project: Analysis of Hierarchical Eigentasks and Change-Point Detection in SRI Lifelong Learning Machine (Co-PI; subcontract with SRI International)
Sponsor: Defense Advanced Research Projects Agency, US Department of Defense
Amount: \$210,000
Duration: 01/01/2021 – 10/31/2021
- Project: Glutamate Neuro-Excitotoxicity in GWI (co-I; PI: Kathleen Holton, American University)
Sponsor: Department of Defense, Gulf War Illness Research Program
Amount: \$914,879
Duration: 09/01/2017 – 08/31/2020
- Project: Statistical methods for analyzing antigen receptors data (Senior personnel; PI: Grzegorz Rempala at Georgia Regents University and the Ohio State University)
Sponsor: National Institutes of Health, Biostatistical Methods and Research Design Study Section (BMRD)
Amount: \$811,957
Duration: 07/01/2010 – 04/30/2015
- Project: Randomized Double-Blind Placebo Controlled Study of PROMETA® (Senior Personnel; PI: Dr. H. Urschel).
Sponsor: Hythiam, Inc.
Amount: Unrestricted
Location: Research Across America, Dallas, TX

3. Grants for Conference Organization

- Project: International Workshop in Sequential Methodologies (PI and Conference Organizer and Chair)
Sponsor: National Science Foundation, Mathematical Sciences
Amount: \$30,000
Duration: 04/01/2026 – 03/31/2027
- Project: Quality and Productivity Research Conference - Data and Science Is a Winning Alliance (PI and Conference Organizer and Chair)
Sponsor: National Science Foundation, Mathematical Sciences
Amount: \$25,655
Duration: 04/01/2019 – 03/31/2020
- Project: The 36-th Quality and Productivity Research Conference, June 10-13, 2019 (PI and Conference Organizer and Chair)
Sponsors: SAS/JMP (\$5000), Amazon (\$5000), Minitab (\$2500), iManage (\$2500), Trilobyte (\$2000), StyleData (\$500)
Total Amount: \$17,500

Books authored and co-authored

1. *Probability and Statistics for Computer Scientists*, by Michael Baron (426 pp.) Chapman & Hall / CRC, Boca Raton, FL, 2007. ISBN 1584886412.
2. Instructor's Solution Manual for *Probability and Statistics for Computer Scientists*, by Michael Baron (99 pp.) Chapman & Hall / CRC, Boca Raton, FL, 2007.
3. *Bourdieu's Demon, Volume 1. Strategies of the Upper Middle Class in the Information Age*, by Richard Baker and Michael Baron (292 pp.) CreateSpace Publishing, North Charleston, SC, 2012. ISBN 147826974X.
4. *Probability and Statistics for Computer Scientists*, by Michael Baron. *Second edition* (473 pp.). Chapman & Hall / CRC, Boca Raton, FL, 2013. ISBN 1439875901.
5. Instructor's Solution Manual for *Probability and Statistics for Computer Scientists*, Second Edition, by Michael Baron (144 pp.) Chapman & Hall / CRC, Boca Raton, FL, 2013.
6. *Bourdieu's Demon, Volume 2. Non-Financial Capital in the 21st Century*, by Richard Baker and Michael Baron (32 pp.) Amazon Digital Services, Inc., Seattle, WA, 2014. ISBN .
7. *Bourdieu's Demon, Volume 3. Premium Knowledge: Marketing Communications*, by Richard Baker and Michael Baron (54 pp.) CreateSpace Publishing, North Charleston, SC, 2014. ISBN 1502722038.

8. *Bourdieu's Demon, Volume 4. Santa Fe Conjecture: Mathematical Proof for the Theory of Moral Sentiments*, by Richard Baker and Michael Baron (24 pp.) CreateSpace Publishing, North Charleston, SC, 2016. ISBN 1518795285.
9. *Bourdieu's Demon, Volume 5. The M Function: The Invisible Hand*, by Richard Baker and Michael Baron (106 pp.) CreateSpace Publishing, North Charleston, SC, 2017. ISBN 1545285705.
10. *Bourdieu's Demon, Volume 6. Theory of the Self and Self-Consciousness*, by Richard Baker and Michael Baron (52 pp.) CreateSpace Publishing, North Charleston, SC, 2018. ISBN 1984052640.
11. *The Science of the Self. Cultural Evolution Through Integration of Biological, Behavioral, Social, and Mathematical Sciences*, by Richard Baker and Michael Baron (404 pp.) CreateSpace Publishing, North Charleston, SC, 2018. ISBN 1718823908.
12. *Probability and Statistics for Computer Scientists*, by Michael Baron. *Third edition* (491 pp.) Chapman & Hall / CRC, Boca Raton, FL, 2019. ISBN 1138044482. [Link](#).
Translated to Chinese and published by China Machine Press in 2023. ISBN 978-7-111-71635-8. [Link](#).
13. *Instructor's Solution Manual for Probability and Statistics for Computer Scientists, Third Edition*, by Michael Baron. Chapman & Hall / CRC, Boca Raton, FL, 2020.
14. *Probability and Statistics for Computer Scientists with R and Python*, by Michael Baron. *Fourth edition*. Chapman & Hall / CRC Press, Boca Raton, FL, 2027 (under contract).

Refereed journal articles and book chapters (* marks supervised students)

1. M. Baron. On the first passage time for waiting processes. *Theory of Probab. and Appl.*, 41: 328 -- 334, 1996. [Link](#).
2. M. Baron and A. L. Rukhin. Asymptotic behavior of confidence regions in the change-point problem. *J. of Stat. Planning and Inference*, 58:263-282, 1997.
3. M. Baron. Convergence rates of change-point estimators and tail probabilities of the first-passage-time process. *Canadian J. of Statistics*, 27:183-197, 1999.
4. M. Baron and A. L. Rukhin. Distribution of the number of visits of a random walk. *Stochastic Models*, 15: 593-597, 1999.
5. M. Baron. Nonparametric adaptive change-point estimation and on-line detection. *Sequential Analysis*, 19 (1-2), 1-23, 2000.
6. M. Baron. On statistical inference under asymmetric loss functions. *Statistics & Risk Modeling*, 18 (4), 367-388, 2000.

7. M. Baron, C. K. Lakshminarayan, and Z. Chen. Markov random fields in pattern recognition for semiconductor manufacturing. *Technometrics*, 43 (1), 66-72, 2001.
8. M. Baron and A. L. Rukhin. Perpetuities and asymptotic change-point analysis. *Statistics and Probability Letters*, 55 (1), 29-38, 2001.
9. M. Baron. Bayes stopping rules in a change-point model with a random hazard rate. *Sequential Analysis*, 20 (3), 147-163, 2001.
10. M. Baron, M. Rosenberg, and N. Sidorenko. Electricity pricing: modeling and prediction with automatic spike detection. *Energy, Power, and Risk Management*, 36-39, October 2001.
11. M. Baron, M. Rosenberg, and N. Sidorenko. Divide and conquer: forecasting power via automatic price regime separation. *Energy, Power, and Risk Management*, 70-73, March 2002.
12. N. Sidorenko, M. Baron, and M. Rosenberg. Estimating oil price volatility: a GARCH model. *Energy, Power, and Risk Management*, 62-65, October 2002.
13. M. Baron. Bayes and asymptotically pointwise optimal stopping rules for the detection of influenza epidemics. In C. Gatsonis, R. E. Kass, A. Carriquiry, A. Gelman, D. Higdon, D. K. Pauler and I. Verdinelli, Eds., *Case Studies in Bayesian Statistics*, vol. 6, pages 153--163, Springer-Verlag, New York, 2002.
14. M. Rosenberg, J. D. Bryngelson, N. Sidorenko; M. Baron. Price spikes and real options: transmission valuation. In E. I. Ronn, ed., *Real Options and Energy Management*, pages 323--370, Risk Books, London, 2002.
15. M. Rosenberg, J. D. Bryngelson; M. Baron. Probability and stochastic calculus: review of probability concepts. In E. I. Ronn, ed., *Real Options and Energy Management*, pages 659--697, Risk Books, London, 2002.
16. M. Baron and N. Granott. Consistent estimation of early and frequent change points. In J. Haitovsky, H. R. Lerche, and Y. Ritov, eds., *Foundations of Statistical Inference*, pages 181--194, Physica-Verlag, Heidelberg, New York, 2003.
17. M. Baron. Sequential methods for multistate processes. In N. Mukhopadhyay, S. Datta, S. Chattopadhyay, eds., *Applied Sequential Methodologies. Real-World Examples with Data Analysis*, 55-73, Marcel Dekker, Inc., New York, 2004.
18. M. Baron. Detection of epidemics as a sequential change-point problem. In V. Antonov, C. Huber, M. Nikulin, V. Polischook, eds.; *Longevity, Aging and Degradation Models in Reliability, Public Health, Medicine and Biology*, volume 2, pages 31--43, St. Petersburg, 2004.
19. R. Gill* and M. Baron. Consistent estimation in generalized broken-line regression. *J. of Stat. Planning and Inference*, 126(2), 441--460, 2004.

20. C. Schmegner* and M. Baron. Principles of optimal sequential planning. *Sequential Analysis*, 23(1), 11--32, 2004.
21. M. Baron and A. Tartakovsky. Asymptotic optimality of change-point detection schemes in general continuous-time models. *Sequential Analysis*, 25(3) 257--296, 2006. DOI: [Link](#).
22. R. Gill*, J. Keating, and M. Baron. Detecting abrupt leaks in blended underground storage tanks. *Communication in Statistics: Theory and Methods*, 35(4) 727--742, 2006.
23. C. Schmegner* and M. Baron. Sequential Plans and Risk Evaluation. *Sequential Analysis*, 26(4), 335--354, 2007.
24. H. C. Urschel, L. L. Hanselka, I. Gromov, L. White, and M. Baron. Open-Label Study of a Proprietary Treatment Program Targeting Type A γ -Aminobutyric Acid Receptor Dysregulation in Methamphetamine Dependence. *Mayo Clinic Proceedings*, 82(10), 1170-1178, 2007.
25. J. W. Cangussu and M. Baron. Automatic Identification of Change Points for the System Testing Process. *COMPSAC* (1), 377--384, 2007.
26. M. Baron, A. Takken, E. Yashchin, and M. Lanzerotti. Modeling and Forecasting of Defect-Limited Yield in Semiconductor Manufacturing. *IEEE Transactions on Semiconductor Manufacturing*, 21(4), 614-624, 2008.
27. H. C. Urschel, L. L. Hanselka, and M. Baron. Drug craving: construct and concurrent validity. *Amer. J. of Addictions*, 17(4), 338, 2008.
28. S. Suzuki* and M. Baron. ϵ -Bayes Sequential Plans. *Advances and Applications in Statistics* 11 (2), 173-183, 2009.
29. M. Rosenberg, J. D. Bryngelson, M. Baron, and A. D. Papalexopoulos. Transmission Valuation Analysis Based on Real Options with Price Spikes, in *Handbook of Power Systems II (Energy Systems)*, S. Rebennack, P.M. Pardalos, M.V.F. Pereira and N. Iliadis (eds). Springer, 2010, pp. 101-125.
30. M. Baron and S. Efromovich. Discussion on "Quickest Detection Problems: Fifty Years Later" by Albert N. Shiryaev, *Sequential Analysis*, 29: 398-403, 2010.
31. J. W. Cangussu, S. W. Haider, K. Cooper, and M. Baron. On the selection of software defect estimation techniques. *Software Testing, Verification and Reliability* 21 (2), 125-152, 2011.
32. H. C. Urschel, L. L. Hanselka, and M. Baron. A controlled trial of flumazenil and gabapentin for initial treatment of methylamphetamine dependence. *J. of Psychopharmacology* 25 (2), 254-262, 2011.
33. S. Suzuki* and M. Baron. Construction of the optimal sequential plan for testing a treatment for an adverse effect, *Sequential Analysis* 30 (3): 261-279, 2011.

34. S. De* and M. Baron. Sequential Bonferroni methods for multiple hypothesis testing with strong control of familywise error rates I and II, *Sequential Analysis* 31 (2), 238-262, 2012.
35. S. De* and M. Baron. Step-up and step-down methods for testing multiple hypotheses in sequential experiments, *J. of Stat. Planning and Inference* 142: 2059-2070, 2012.
36. M. Baron. Discussion on “Change-points: From Sequential Detection to Biology and Back” by David O. Siegmund, *Sequential Analysis* 32 (1), 15-18, 2013.
37. X. Yu*, M. Baron, and P. Choudhary. Change-Point Detection in Binomial Thinning Processes, with Applications in Epidemiology, *Sequential Analysis* 32 (3), 350-367, 2013.
38. C. K. Lakshminarayan and M. Baron. Pattern recognition in large-scale data sets: application in integrated circuit manufacturing. In V. Bhatnagar and S. Srinivasa, eds., *Big Data Analytics*, 185-196, Springer, Heidelberg, Germany, 2013.
39. X. Chen* and M. Baron. Change-point analysis of survival data with application in clinical trials, *Open J. of Statistics* 4, 663-677, 2014. DOI: [Link](#).
40. M. Baron. Asymptotically pointwise optimal change detection in multiple channels, *Sequential Analysis* 33 (4), 440-457, 2014.
41. A. Haque*, S. Chandra, L. Khan, and M. Baron. MapReduce guided approximate inference over graphical models. In *2014 IEEE Symposium on Computational Intelligence and Data Mining (CIDM)*, 446-453, IEEE, 2014. DOI: [Link](#).
42. A. Mustafa*, A. Haque*, L. Khan, M. Baron, and B. Thuraisingham. Evolving stream classification using change detection. In *10th IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom)*, 154—162, IEEE, 2014. DOI: [Link](#).
43. S. De* and M. Baron. Sequential tests controlling generalized familywise error rates, *Statistical Methodology* 23, 88-102, 2015.
44. A. Haque*, L. Khan, and M. Baron. Semi-supervised adaptive framework for classifying evolving data stream. In T. Cao, E.-P. Lim, Z.-H. Zhou, T.-B. Ho, D. Cheung, and H. Motoda, eds., *Advances in Knowledge Discovery and Data Mining, Part II*, 383—394, Cham, Switzerland, 2015.
45. T. Zhao* and M. Baron. Multiple tests in group sequential clinical trials, *Jacobs Journal of Biostatistics* 1 (1):004, 2016.
46. M. Baron and S. Zacks. Discussion on “Sequential detection/isolation of abrupt changes” by Igor V. Nikiforov. *Sequential Analysis* 35 (3), 302-304, 2016.
47. A. Haque*, L. Khan, and M. Baron. SAND: Semi Supervised Adaptive Novel Class Detection and Classification over Data Stream. *Proc. of Thirtieth AAAI Conference on Artificial Intelligence (AAAI-16)*, 1652—1658, Phoenix, AZ, 2016.

48. A. Haque*, L. Khan., M. Baron, B. Thuraisingham, and C. Aggarwal. Efficient handling of concept drift and concept evolution over stream data. *Proc. of 2016 IEEE 32nd International Conference on Data Engineering (ICDE)*, 481-492, 2016. [Link](#).
49. M. Baron and R. Xu*. Sequential testing for full credibility, *Variance* 10 (2), 227-239, 2018. [Link](#).
50. M. Baron and L. MacMillan*. Weighted Statistics for Testing Multiple Endpoints in Clinical Trials, *Annals of Biostatistics and Biometric Applications* 2(2): 2019.
51. K. F. Holton, A. E. Kirkland*, M. Baron, S. S. Ramachandra*, M. T. Langan, E. T. Brandley, and J. N. Baraniuk. The Low Glutamate Diet Effectively Improves Pain and Other Symptoms of Gulf War Illness, *Nutrients*, 12(9):2593, 2020. DOI: [Link](#).
52. M. Baron and E. Yashchin. Foreword: Special Issue on Statistics in Quality and Productivity. *Applied Stochastic Models in Business and Industry*, 36 (6): 977-979, 2020. [Link](#).
53. M. Baron, S. Ye*, and B. Chattopadhyay. Limited-fluctuation credibility under uncertain priors, *Variance*, 14 (1): 2021. [Link](#).
54. K. F. Holton, S. S. Ramachandra*, S. L. Murray*, M. Baron, and J. N. Baraniuk. Effect of the Low Glutamate Diet on Inflammatory Cytokines in Veterans with Gulf War Illness (GWI): A Pilot Study. *Life Sciences*, 280, 119637: September 2021. [Link](#).
55. K. Faber, R. Corizzo, B. Sniezynski, M. Baron and N. Japkowicz. WATCH: Wasserstein Change Point Detection for High-Dimensional Time Series Data. *IEEE International Conference on Big Data*, pp. 4450-4459, 2021. [Link](#).
56. A. E. Kirkland*, M. Baron, J. W. VanMeter, J. N. Baraniuk, and K. F. Holton. The Low Glutamate Diet Improves Cognitive Functioning in Veterans with Gulf War Illness and Resting-State EEG Potentially Predicts Response. *Nutritional Neuroscience*, 25 (11): 2247-2258, 2022. DOI: [Link](#).
57. E. T. Brandley, A. E. Kirkland*, M. Baron, J. N. Baraniuk, and K. F. Holton. The Effect of the Low Glutamate Diet on the Reduction of Psychiatric Symptoms in Veterans With Gulf War Illness: A Pilot Randomized-Controlled Trial. *Frontiers in Psychiatry*, Sec. Mood Disorders, 13, June 2022. [Link](#).
58. M. Baron, E. Yashchin, and A. Takken. Introduction to Wafer Tomography: Likelihood-Based Prediction of Integrated-Circuit Yield. In *Artificial Intelligence, Big Data and Data Science in Statistics*, edited by A. Steland and K.-L. Tsui, pp. 227-252. Springer, 2022. [Link](#).
59. R. Corizzo, M. Baron, and N. Japkowicz. CPDGA: Change Point Driven Growing Auto-Encoder for Lifelong Anomaly Detection. *Knowledge Based Systems*, 247, July 2022. [Link](#).
60. K. Faber, R. Corizzo, B. Śnieżyński, M. Baron, and N. Japkowicz. LIFEWATCH: Lifelong Wasserstein Change Point Detection. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pp. 1-8. IEEE, 2022.

61. I. Sur, Z. Daniels, A. Rahman, K. Faber, G. Gallardo, T. Hayes, C. Taylor, M. B. Gurbuz, J. Smith, S. Joshi, N. Japkowicz, M. Baron, Z. Kira, C. Kanan, R. Corizzo, A. Divakaran, M. Piacentino, J. Hostetler, and A. Raghavan. System design for an integrated lifelong reinforcement learning agent for real-time strategy games. In *Proceedings of the Second International Conference on AI-ML Systems*, article 12, pp. 1—9, 2022. DOI: [Link](#).
62. M. Baron and S. V. Malov. Detection and Estimation of Multiple Transient Changes. *Journal of Applied Statistics*, 50 (14) pp. 2862-2888, 2023, DOI: [Link](#).
63. M. Baron and S. D. Casey. Sampling Expansions for AFB Signals via Change Point Detection. 2025 International Conference on Sampling Theory and Applications (SampTA), Vienna, Austria, pp. 1-5, IEEE, 2025. DOI: [Link](#).
64. M. Baron and S. V. Malov. Detection of Temporary Disorders with Application to Electricity Pricing. In: S. Kovalev, I. Kotenko, A. Sukhanov (eds), *Intelligent Information Technologies for Industry (IITI'25)*, Vol. 1. Lecture Notes in Networks and Systems, vol. 1762. Springer, Cham. DOI: [Link](#).
65. M. Baron and Y. Malinovsky. The 9th International Workshop in Sequential Methodologies. *Sequential Analysis*. Accepted for publication. DOI: [Link](#).

Under review

66. F. Martami, A. Maury, H. Taheri, N. A. Jafari, S. Murray, M. Baron, and K. F. Holton. Confirmation of the Low Glutamate Diet as an Effective “Food is Medicine” Treatment for Chronic Multisymptom Neurological Illness in Veterans.

In preparation

67. S. Malov and M. Baron. Practical properties of the CUSUM Process. *Communications in Statistics*, special issue; to be submitted.
68. M. Baron and S. Malov. Detection of Genome Instability Regions Using Coverage Signal; to be submitted.
69. M. Baron. Incomplete data and estimation of under-reported counts in the COVID-19 pandemic. In preparation.
70. M. Baron and A. Cameron. Minimax Methods for Multiple Hypothesis Testing, to be submitted.

Invited presentations

1. On Bayesian inference under non-symmetric loss functions. Statistics Colloquium at Texas AM University, College Station TX, November 2, 1995

2. Nonparametric detecting abrupt distributional changes in sequentially obtained data. 3rd IMS New Researchers Conference, Laramie WY, July 1997 .
3. Statistical problems of an asymmetric nature. Meeting of AMS, Louisville, KY, March 1998 .
4. New developments in nonparametric change-point analysis. Academic seminar at the University of Louisville. Louisville KY, 10/16/98.
5. Change-point estimation, sequential and retrospective procedures, and global warming. Academic seminar at Texas A & M University. Commerce TX, 11/03/98.
6. Detection of multiple change points: on-line and off-line algorithms. Invited talk. Statistics seminar at the University of Maryland, Baltimore County. Baltimore MD, 11/20/98.
7. Bayesian and minimax estimation under non-standard loss functions. The 4th New Researchers Conference in Statistics. Baltimore, MD, August 1999.
8. Consistent estimation of early and frequent change points in processes of learning and development. The Foundations of Statistical Inference, Jerusalem, Israel, December 2000.
9. Fundamentals and new developments in statistical change-point analysis. Seminar at TXU Energy Trading, Dallas TX, 02/23/2001 .
10. Markov random fields in semiconductor manufacturing. Technometrics invited paper. Spring Research Conference on Statistics in Industry and Technology, Roanoke, VA, June 2001.
11. Optimal stopping rules for change-point detection. The 8th International Vilnius Conference on Probability Theory and Mathematical Statistics, Vilnius, Lithuania, June 2002.
12. Optimal stopping rules in sequential Bayesian change-point detection problems. Probability Seminar Series at Columbia University, New York, 11/01/02 .
13. Divide and conquer: forecasting power markets via automatic price regime separation. POWER-GEN International Conference, Orlando, FL, December 2002 .
14. Sequential Plans: Principles and Applications. International Conference on Ranking and Selection, Multiple Comparisons, Reliability, and Their Applications, Chennai, India, December 2002.
15. Asymptotic Analysis of Bayesian Quickest Change Detection Procedures. Quality Productivity Research Conference, IBM T. J. Watson Research Ctr., Yorktown Heights, NY, May 2003.
16. Optimizing Sequential Design of Experiments. Special invited paper. Quality & Productivity Research Conference, IBM T. J. Watson Research Ctr., Yorktown Heights, NY, May 2003.
17. Sequential change-point analysis for the early detection of epidemics. Invited tutorial lecture. DIMACS Tutorial on Statistical and Other Analytic Health Surveillance Methods, Special Focus on Computational and Mathematical Epidemiology, DIMACS Center, Rutgers University, Piscataway, NJ, June 2003 .

18. Spatial modelling and estimation in semiconductor manufacturing. The 1st Joint Statistical Meeting of IMS and ISBA, San Juan, Puerto Rico, July 2003 .
19. Optimal Sequential Detection of Change-Points and Bayesian Analysis of Markov Chains. Department of Statistics Colloquia Series, U. of Connecticut, Storrs, CT, 10/22/03 .
20. Optimal stopping rules in Bayesian change-point detection. Invited talk. Department of Statistics Colloquia Series, New York University, Stern School of Business, New York, 01/23/04 .
21. Hidden Markov processes in energy pricing and option valuation. Fifth Biennial International Conference on Statistics, Probability and Related Areas, Athens GA, May 2004 .
22. Statistical methods of early detection of epidemics. The First International French-Russian Workshop "Longevity, Aging and Degradation Models in Reliability, Public Health, Medicine and Biology", St.Petersburg, Russia, June 2004.
23. Defects Modeling and Yield Forecasting in Semiconductor Manufacturing. Sematech Symposium, Austin, TX, October 2004
24. Optimization of sequential plans for clinical trials. 25th European Meeting of Statisticians, Oslo, Norway, August 2005
25. Construction of Optimal Sequential Plans. First International Workshop in Sequential Methodologies (IWSM 2007), Auburn AL, July 2007
26. Drug Craving: Construct and Concurrent Validity (with H. C. Urschel and L. L. Hanselka). The 18th Annual Meeting and Symposium of American Academy of Addiction Psychiatry (AAAP), Coronado CA, November-December 2007
27. Construction of Optimal Sequential Plans. The 2nd Annual Probability and Statistics Day at UMBC, Baltimore MD, April 2008.
28. A Controlled Trial of Flumazenil and Gabapentin for Initial Treatment for Methamphetamine Dependence (with H. Urschel and L. Hanselka). Presented at the 70th Annual Meetings of the College on Problems of Drug Dependence in San Juan PR, June 18, 2008.
29. Detection of Epidemics as a Sequential Change-Point Problem. The 3rd Annual Probability and Statistics Day at UMBC, Baltimore MD, April 2009.
30. Optimization of error spending and power spending in sequential designs. Invited Paper. The 4th Annual Probability and Statistics Day at UMBC, Baltimore MD, April 2010.
31. Detection of leaks in underground storage tanks as a multiple change-point problem. Invited talk at the Louisiana Chapter of ASA. Lafayette LA, November 2010.

32. Sequential methods for multiple hypotheses testing with strong control of Type I and Type II familywise error rates (with S. K. De*). Invited paper at the Third International Workshop in Sequential Methodologies. Stanford University, Palo Alto CA, June 2011.
33. Construction of an optimal sequential plan for testing a treatment for an adverse effect (with S. Suzuki*). Invited paper at the Third International Workshop in Sequential Methodologies. Stanford University, Palo Alto CA, June 2011.
34. Asymptotically efficient sequential tests of multiple hypotheses (with S. K. De*). Joint Statistical Meetings. Miami Beach FL, August 2011.
35. Sequential change-point detection in the spread of infectious diseases. Invited paper at the Joint Statistical Meetings. Miami Beach FL, August 2011.
36. Testing multiple hypotheses in sequential experiments. Biostatistics Colloquium at Georgia Health Sciences University. Augusta GA, February 2012.
37. Multiple hypotheses testing in sequential experiments. Mathematics and Statistics seminar series at the University of Texas at El Paso, March 2012.
38. Stepwise methods of multiple hypotheses testing in sequential experiments, presented by S. De*. Poster at the Conference of Texas Statisticians (First Place in the Poster Session Award), Beaumont TX, March 2012.
39. Multiple hypotheses testing in sequential experiments. Invited seminar at the University of Arkansas, Little Rock AR, March 2012.
40. Change-point detection in multiple channels. Invited paper at the Fourth International Workshop in Sequential Methodologies. University of Georgia, Athens GA, July 2013.
41. Sequential multiple testing controlling generalized error rates (with S. De*). Invited paper at the Fourth International Workshop in Sequential Methodologies. University of Georgia, Athens GA, July 2013.
42. Multiple testing in group sequential clinical trials (with T. Zhao*). Invited paper at the Fourth International Workshop in Sequential Methodologies. University of Georgia, Athens GA, July 2013.
43. Change-point detection with multiple sensors (with T. Wang*). Invited paper at the Fourth International Workshop in Sequential Methodologies. University of Georgia, Athens GA, July 2013.
44. Asymptotically pointwise optimal stopping rules for multichannel change-point detection. Invited seminar at Binghamton University – State University of New York, Binghamton NY, October 2013.

45. Pattern recognition in large-scale data sets: application in integrated-circuit manufacturing (with C. K. Lakshminarayan). Second International Conference on Big Data Analytics. Mysore, India, December 2013.
46. Use of prior information for threat detection with single and multiple sensors. Algorithms for Threat Detection Program Review. Boulder CO, March 2014.
47. Multi-sensor change-point detection schemes (with T. Wang). Algorithms for Threat Detection Program Review. Boulder CO, March 2014.
48. Multidimensional nonparametric change-point detection using data mining in security domain (with A. Mustafa, A. Haque, L. Khan, and T. Wang). Algorithms for Threat Detection Program Review. Boulder CO, March 2014.
49. Change-point detection in multivariate time series. Recent Advances and Trends in Time Series Analysis, Nonlinear Time Series, High Dimensional Inference, and Beyond. Banff AB, Canada, May 2014.
50. Controlling familywise error rate and power in sequential experiments. 2014 IMPACT Symposium III "Advances in Clinical trial Statistics: Multiplicity Adjustment and Sequential, Multiple Assignment, Randomized Trials". Cary NC, November 2014.
51. Optimal sequential planning of clinical trials. Invited colloquium presentation. Johns Hopkins University, Baltimore MD, April 2015.
52. Enhanced stepwise methods for multiple comparisons in sequential experiments. Invited paper at the Fifth International Workshop in Sequential Methodologies. Columbia University, New York, June 2015.
53. Statistical control of sensitivity and accuracy in multi-sensor detection of potential threats (with L. Khan, T. Wang, A. Mustafa, and A. Haque). Algorithms for Threat Detection Program Review. NSF, Arlington VA, July 2015.
54. Classification of Evolving Data Streams using Dynamic Chunk Size and Limited Labeled Data (with L. Khan and A. Haque). Algorithms for Threat Detection Program Review. NSF, Arlington VA, July 2015.
55. Cost reduction techniques in multiple hypothesis testing. International Workshop on Applied Probability (IWAP). Toronto ON, Canada, June 2016.
56. Wafer Tomography: Study of Defects and Prediction of Integrated-Circuit Yield (with E. Yashchin and A. Takken). Joint Statistical Meetings (JSM), Chicago IL, August 2016.

57. Weighted statistics, minimax error spending, and other cost reduction techniques in multiple hypothesis testing. Statistics seminar at the University of Maryland, Baltimore County. Baltimore MD, September 2016.
58. Testing Multiple Hypotheses in Sequential Experiments. The Biostatistics Center, Milken Institute School of Public Health, The George Washington University, Rockville MD, December 2016.
59. Analysis of big, high-dimensional, and mostly hidden data for semiconductor yield prediction. The 4th African International Conference on Statistics. Polokwane, South Africa, March 2017.
60. Simultaneous quality control in multiple sequences. Quality and Productivity Research Conference, Storrs CT, June 2017.
61. Change-Point Detection in Binomial Thinning Processes, with Applications in Epidemiology.. International Workshop in Sequential Methodologies. Rouen, France, June 2017.
62. Statistical Algorithms for Real-Time Detection of Changes in Social Patterns. Algorithms of Threat Detection Kickoff Workshop, Washington DC, September 2017.
63. Multiple hypothesis testing in sequential clinical trials. National Institutes of Health, Office of Biostatistics Research Seminar, Bethesda MD, February 27, 2018.
64. Statistical Change Detection Methods. Zero Project Workshop. National Strategic Research Institute facility, Annapolis Junction MD, April 11-12, 2018.
65. Bayesian-type change-point detection in statistical process control. Joint Research Conference the Spring Research Conference (SRC) and the Quality and Productivity Research Conference (QPRC). Santa Fe NM, June 2018.
66. Decision theoretic approach to sequential testing of multiple hypotheses. The 9th International Workshop on Applied Probability (IWAP). Budapest, Hungary, June 2018.
67. Stepwise, minimax, and weighted methods for sequential tests of multiple hypotheses. The 27th Nordic Conference in Mathematical Statistics (Nordstat). Tartu, Estonia, June 2018.
68. Optimizing sequential testing of multiple hypotheses. The 12th International Vilnius Conference on Probability Theory and Mathematical Statistics and 2018 IMS Annual Meeting on Probability and Statistics. Vilnius, Lithuania, July 2018.
69. Detection of patterns in geospatial sequences of social events. Algorithms of Threat Detection Workshop, Washington DC, October 2018.

70. Change-Point Detection in a Markov Random Field. The 32nd European Meeting of Statisticians. Palermo, Italy, July 2019.
71. Detection of changes in spatial data. The Joint Statistical Meetings. Denver CO, July-August 2019.
72. Growing Hierarchical Auto-Encoder (with R. Corizzo and N. Japkowicz). Lifetime Learning Machines PI Review. Chicago IL, September 2019.
73. Sequential testing and post-analysis of credibility. International Conference on Statistical Distributions and Applications (ICOSDA). Grand Rapids MI, October 2019.
74. Spatio-temporal modeling and change-point analysis of social events. Algorithms of Threat Detection Workshop, Washington DC, October 2019.
75. SIR models, thinning processes, and change-points: analysis of epidemic counts data. Mathematics and Statistics Colloquium, American University, Washington DC, June 2020.
76. Modeling and estimation for COVID-19 under-reported counts (with E. Baron). Data Science Conference on COVID-19 [DSCC-19]. Virtual, August 2020.
77. Statistical analysis of epidemic counts data: modeling, detection of outbreaks, and recovery of missing data. Mathematics and Statistics Colloquium, American University, Washington DC, September 2020.
78. Change-point detection in the correlation structure of time series. District Fourier Talks (DFT), virtual by American University, October 2020.
79. Estimation of under-reported epidemic counts. Mathematical Association of America, MD-DC-VA Section, (virtual), November 2020.
80. Sequential Analysis of Epidemic Threats. Algorithms of Threat Detection Workshop (virtual), November 2020.
81. Wafer Tomography and Quality Control in Semiconductor Manufacturing. International Virtual Conference on Advanced Statistical Techniques in Business and Industry (ISBIS). Cochin, Kerala, India (virtual), December 2020.
82. Incomplete data in the COVID-19 pandemic. The 63rd ISI World Statistics Congress, The Hague, Netherlands, <https://www.isi2021.org/> (virtual), July 2021.
83. Detection of temporary disorders. Bernoulli World Congress, Seoul, Korea, <https://www.wc2020.org/> (virtual), July 2021.

84. Detection of Temporary System Failures. Joint Statistical Meetings, Seattle WA, <https://ww2.amstat.org/meetings/jsm/2021/> (virtual), August 2021.
85. Limited-Fluctuation Credibility under Uncertain Priors. The 56th Actuarial Research Conference, Chicago IL (virtual), August 2021.
86. Detection of multiple transient changes. International Chinese Statistical Association (ICSA) Applied Statistics Symposium, <https://www.icsa.org/category/events/> (virtual), September 2021.
87. Sequential estimation of under-reported epidemic counts (with E. Baron). The 34th New England Statistics Symposium (NESS), Providence RI, October 2021.
88. Yield Modeling for Multi-Layer Products (with E. Yashchin and A. Takken, presented by E. Yashchin). INFORMS Annual Meeting (virtual), Anaheim CA, October 2021.
89. Detection of transient changes and application in statistical genetics (with S. Malov). IMS Annual Meeting in Probability and Statistics, London UK, June 27-30, 2022.
90. Detection of transient changes and application in energy finance (with S. Malov). The Joint Statistical Meetings, Washington DC, August 6-11, 2022.
91. Detection and Estimation of Multiple Transient Changes (with S. Malov). XIV-th International Workshop on Intelligent Statistical Quality Control, Washington DC, August 15-17, 2023
92. Where Data Meets Math Sciences: A Conversation with Dr. Michael Baron. STEM Time seminar, American University, Washington DC, September 27, 2023.
93. Multiple testing in sequential experiments. Academic seminar at the Division of Biostatistics, Medical College of Wisconsin, Milwaukee WI, October 24, 2023.
94. Multiple testing in sequential experiments. Plenary talk at the 8th International Workshop in Sequential Methodologies, Orem UT, May 21-24, 2024.
95. Weighting and Equalizer Methods for Risk Optimization in Multiple Testing. Eleventh International Triennial Calcutta Symposium on Probability and Statistics, Kolkata, India, December 27-30, 2024.
96. Sampling Expansions for AFB Signals via Change Point Detection (with S. Casey). International Conference on Sampling Theory and Applications (SampTA), Vienna, Austria, July 28 – August 1, 2025.
97. Minimax Methods in Multiple Testing. The 13th International Conference on Multiple Comparison Procedures (MCP–2025). Temple University, Philadelphia PA, August 12–15, 2025.

98. Bayesian and Asymptotically Pointwise Optimal Detection of Changes in Multiple Channels. Academic seminar at the U.S. Naval Academy, Mathematics Department, Annapolis MD, October 22, 2025.
99. Multiple Testing in Sequential and Non-Sequential Experiments. Invited talk at the Austin TX Chapter of the American Statistical Association. November 19, 2025.
100. Multiple Testing in Sequential and Non-Sequential Clinical Trials. Invited talk at Basel Biometric Society (BBS), Austrian-Swiss region. March 9, 2026.
101. Multiple Testing and Minimax Error Spending. StatConnect@AI conference, Georgetown University, Washington, DC, March 27, 2026.
102. Sampling Expansions for AFB Signals via Change-Point Detection (with S. Casey). The 9th International Workshop in Sequential Methodologies. American University, Washington, DC, June 1-4, 2026.
103. Transient Change-Point Detection via Finite-Sample Properties of the CUSUM Process. The 9th International Workshop in Sequential Methodologies. American University, Washington, DC, June 1-4, 2026.
104. Minimax Error Spending and Cost Saving in Multiple Testing (with A. Cameron and S. Malov). Joint Statistical Meetings, Boston, MA, August 1-6, 2026.

Research Supervision

Supervised Doctoral Dissertations

1. Ryan Gill

Dissertation: Introduction to generalized broken line regression

Graduation: May 2002

Currently: Professor of Statistics with tenure at the University of Louisville, KY

2. Claudia Schmegner

Dissertation: Decision theoretic results for sequentially planned statistical procedures

Graduation: August 2003

Currently: Associate Professor of Statistics with tenure at DePaul University, Chicago, IL

3. Sumihiro Suzuki

Dissertation: Constructive methodologies of optimal sequential plans

Graduation: December 2006

Currently: Professor and Director of Biostatistics and Epidemiology, Department of Preventive Medicine, Rush University, Chicago IL

Formerly: Associate Professor of Statistics and Chair of the Department of Biostatistics and Epidemiology at the University of North Texas, School of Public Health, Fort Worth, TX

4. Jingsi Xia

Dissertation: Optimal sequentially planned change-point detection procedures

Graduation: December 2006

Currently: Senior Manager for Fraud Risk at Discover Financial Services, Houston, TX

5. Xuan Chen

Dissertation: Change-point analysis of survival data with application in clinical trials

Graduation: August 2009

Currently: Senior Statistician in the Biostatistics Research Group, Sanofi Pasteur, Beijing, China

6. Yi Zhong

Dissertation: Optimization of Error Spending and Power Spending in Sequentially Planned Statistical Experiments

Graduation: December 2010

Currently: Institute Research Investigator at the University of Texas MD Anderson Cancer Center, Houston, TX

7. Xian Yu (co-advisor: Dr. Pankaj Choudhary)

Dissertation: Sequential Change-Point Analysis of Markov Chains with Application to Fast Detection of Epidemic Trends

Graduation: December 2010

Currently: Sr. Biostatistician, Methodology & Analytics Core, Center for Innovations in Quality, Effectiveness and Safety, Michael E. DeBakey VA Medical Center, Houston, TX

Formerly: Sr. Biostatistician, Department of Medicine, Baylor College of Medicine, Houston, TX
Assistant Professor of Statistics at the University of Arkansas, Department of Mathematics and Statistics, Little Rock, AR

8. Rui Xu

Dissertation: Sequential analysis of credibility and insurance risks

Graduation: May 2012

Currently: Statistical Analyst at Axxess Integrate, Dallas TX

9. Shyamal De

Dissertation: Simultaneous Testing of Multiple Hypotheses in Sequential Experiments

First place at the Conference of Texas Statisticians (COTS-2012) with a monetary award

Graduation: August 2012

Currently: Assistant Professor, Applied Statistics Division, Indian Statistical Institute, Kolkata, India
Formerly: Assistant Professor of Statistics at the National Institute for Science Education and Research, Bhubaneswar, OD, India

10. Tian Zhao

Dissertation: Multiple Comparisons in Truncated Sequential Experiments, with Applications in Clinical Trials
Graduation: May 2015
Currently: Senior Scientist at Merck & Co, Rahway NJ

11. Tiansong Wang

Dissertation: Multi-Sensor Change-point Detection
Graduation: August 2015
Currently: Statistician at GMG Holdings, Columbus OH

12. Cesar Alfonso Jamie Acosta-Mejia

Dissertation: Pseudolikelihood Methods in Multichannel Change-Point Detection
Graduation: Summer 2016
Currently: Senior Lecturer at the University of Southern California, Los Angeles CA

Supervised M.S. Theses

1. Kathryn Bean

Thesis: Evaluation and modification of clustering algorithms with missing data
Graduation: August 2008
Continued in Ph.D. program in Computer Science at UT-Dallas

2. Cesar Acosta-Mejia

Thesis: Pseudolikelihood based approach to change-point detection in multiple channels
Graduation: May 2013
Continued in Ph.D. program in Statistics at UT-Dallas

3. John Schuler

Thesis: The Effect of Monetary Injections on Price Variance
Graduation: May 2017
Continued in Ph.D. program in Economics at George Mason University

4. Zachary Nadrich

Thesis: A Spatial-Temporal Markov Random Field Model.
Graduation: December 2020

Supervised B.S. Honors Theses and Honors Capstone Projects

1. Giang David Nguyen

Thesis: Estimating change-points in a multiparameter exponential family

Graduation: December 1997

Employed in J. C. Penney Insurance Group, Plano TX

2. Byron Campen

Thesis: An investigation of the arrival process of unsolicited email

Graduation: December 2004

Continued to M.S. program in Computer Science at UT-Dallas

3. Shangyuan Ye

Thesis: Statistical methods for gaining precision in insurance and business

Winner of the 2013 Undergraduate Research Scholarship Award for this project.

Graduation: May 2014

Continued to M.S. and Ph.D. programs in Statistics at the University of Oklahoma. After postdoctoral experience at Harvard Medical School and Harvard Pilgrim Health Care Institute, became an assistant staff scientist at the Oregon Health and Science University.

4. Emily Hanson (at American University)

Thesis: Performance evaluation and comparison of sequential multiarm bandit strategies.

Graduation: May 2016

Supervised High School Students

1. Jeff Delezen (summer research at UT-Dallas sponsored by Clarke Foundation)

Topic: Regression Analysis of Economic and Real Estate Data

Semester: Summer 1996

After graduation from high school, continued in Statistics at Texas A & M University

Teaching and Curriculum Development

Courses and other curricula developed

(* marks the courses introduced and developed by M. Baron)

Undergraduate courses

STAT 202 Basic Statistics (at AU)

STAT 204* Introduction to Business Statistics (at AU)

STAT 320	Biostatistics (at AU)
STAT 415	Regression (at AU)
STAT 418*	Bayesian Statistics (at AU)
STAT 422	Advanced Biostatistics (at AU)
STAT 424/427*	Data Analysis / Statistical Machine Learning (at AU)
MATH 1314	College Algebra (at UTD)
STAT 1342	Statistical Decision Making (at UTD)
MATH 2418	Linear Algebra (at UTD)
CS/SE/STAT 3341*	Probability and Statistics for Computer Science and Software Engineering (UTD)
EE 3341	Probability and Statistics in Engineering (at UTD)
STAT 4372*	Actuarial Mathematics (at UTD)
MATH 151	Calculus I (at UMBC)
MATH 470	Actuarial Science (at UMBC)
STAT 355	Probability and Statistics for Science and Engineers (at UMBC)

Graduate service courses

STAT 5311	Applied Statistics for Management Sciences (at UTD)
MATH 5390*	Probability and Stochastic Processes in CS (at UTD)

MS level courses

STAT 517*	<i>Special Topics: Decision Theory and Bayesian Inference</i> (at AU)
STAT 517*	<i>Special Topics: Statistical Methods in Clinical Trials</i> (at AU)
STAT 520	Applied Multivariate Analysis (at AU)
STAT 615	Regression (at AU)
STAT 618*	Bayesian Statistics (at AU)
STAT 622	Advanced Biostatistics (at AU)
STAT 624/627*	Data Analysis /Statistical Machine Learning (at AU)
STAT 630	Mathematical Statistics I (at AU)
STAT 631	Mathematical Statistics II (at AU)
DATA 645*	Neural Networks and Deep Learning (online version at AU)
STAT 6326	Sampling Theory (at UTD)
STAT 6337	Advanced Statistical Methods I (at UTD)
STAT 6338	Advanced Statistical Methods II (at UTD)
STAT 6344	Probability Theory I (at UTD)
STAT 6365	Statistical Quality and Process Control (at UTD)
STAT 6390/8303*	<i>Topics in Statistics: Actuarial Mathematics</i> (at UTD)
STAT 6390*	<i>Topics in Statistics: Sequential Methods in Clinical Trials</i> (at UTD)

PhD level courses

STAT 7330	Decision Theory and Bayesian Analysis (at UTD)
STAT 7338	Time Series Modeling and Filtering (at UTD)
STAT 7345	Advanced Probability and Stochastic Processes (at UTD)

STAT 7390*	<i>Topics in Statistics: Sequential Analysis (at UTD)</i>
STAT 8303*	<i>Topics in Statistics: Change-Point Analysis and Related Methods (at UTD)</i>
STAT 8303*	<i>Topics in Statistics: Sequential Methods in (at UTD)</i>

Developed the Biostatistics track within the MS program in Statistics.

Developed the Actuarial Science emphasis within the Statistics undergraduate major.

Developed graduate degree plans for Statistics, Biostatistics, and Quantitative Analysis graduate students.

Participated in the development of new graduate programs in Biostatistics, Data Science, Mathematics of Information and Security, Mathematics for AI.

Developed undergraduate degree plans for Statistics undergraduate students.

Developed Experimental Calculus recitation with Maple (at UMBC).

Supervised independent studies, graduate and undergraduate research, capstone projects.

Supervised 12 Ph.D. dissertations, 4 M.S. theses, 4 B.S. honors theses, and 1 high school student.

Professional Leadership in Statistics

Conference organization

- Organizer, Chair, and Host of the [Conference](#): “The 9th International Workshop in Sequential Methodologies” (IWSM), American University, Washington DC, June 1–4, 2026 (183 participants).
- Organizer, Chair, and Host of the [Conference](#): “The XIVth International Workshop on Intelligent Statistical Quality Control” (ISQC), American University, Washington DC, August 15-18, 2023.
- Organizer, Chair, and Host of the [Conference](#): “The 36-th Quality and Productivity Research Conference. Data and Science is a Winning Alliance”, American University, Washington DC, June 10-13, 2018 (135 participants, including 47 students)
- Organizer and Chair of the Conference: “Algorithms of Threat Detection and Algorithms for Modern Power Systems Joint Workshop”, American University, Washington DC, October 10-12, 2018 (~130 participants)
- Organizer and Chair of the Conference: “Algorithms of Threat Detection Kickoff Workshop”, American University, Washington DC, September 14-15, 2017 (~80 participants)
- Organizer and Chair of the Conference: “The First UT-Dallas Symposium in Statistics and Biostatistics”, UT-Dallas, April 30, 2005

Other conference committee work

- Member of the International Program Committee for the 3rd International Workshop for Sequential Methodologies, Stanford University, Palo Alto, CA, 2011
- Member of the International Program Committee for the 4th International Workshop for Sequential Methodologies, University of Georgia, Athens, GA, 2013
- Member of the International Program Committee for the 8th International Workshop for Sequential Methodologies, Utah Valley University, Orem, UT, 2024

- Member of the International Program Committee for the 9th International Workshop for Sequential Methodologies, American University, Washington, DC, 2026
- Co-organizer of six invited paper sessions, “Sequential Methods for Signals and Complex Data”, “Adaptive Designs and Clinical Trials”, “Optimality in Sequential Estimation and Testing”, “Statistical Learning and High-Dimensional Inference”, “Sequential Monitoring and Detection”, and “Sequential Modeling and Applications” at the 9th International Workshop for Sequential Methodologies, American University, Washington, DC, 2026
- Organizer of four invited paper sessions, “Sequential Methods for Correlated Data”, “Advances in Change-Point Detection and Applications”, “Sequential Methods for Modern Applications”, and “Sequential Methods for Multivariate Data” at the 8th International Workshop for Sequential Methodologies, Utah Valley University, Orem UT, 2024
- Organizer of two invited paper sessions, “Sequential Methods in Quality Control” and “Frontiers in Applied Change-Point Detection” at the 2017 Quality and Productivity Research Conference, University of Connecticut, Storrs, CT
- Organizer of the invited paper session “Multiple Comparisons in Sequential Experiments” at the 4th International Workshop on Sequential Methodologies, University of Georgia, Athens, GA, 2013
- Organizer of the invited paper session “Sequential Designs in Clinical Trials” at the 2011 International Workshop on Sequential Methodologies, Stanford University, Palo Alto, CA
- Organizer of the Invited paper session “Sequential Methods” at 2003 Quality Productivity Research Conference, Yorktown Heights, NY
- Organizer of the Special contributed paper session “Statistical Methods in Semiconductor Manufacturing” at the Joint Statistical Meetings, Dallas TX, 1998
- Member of the Advisory Committee, the Twelfth International Triennial Calcutta Symposium on Probability & Statistics, University of Calcutta, December 28-31, 2027
- Member of the Conference of Texas Statisticians Program and Local Arrangements Committee, 2013-2014
- Member of the International Advisory Committee for the International Conference on Ordered Data Analysis, Models, and Health Research Methods, 2013-2014
- Member of the Quality and Productivity Research Conference committee.
- Chair of numerous invited, special contributed, and contributed sessions.

Editorial work

- Guest Editor of Applied Stochastic Models in Business and Industry
- Associate Editor of the Journal of Sequential Analysis
- Associate Editor of the Journal Applied Stochastic Models in Business and Industry

Refereeing and reviewing

- Regular panel service at the National Science Foundation

- Referee for the Annals of Statistics, Journal of American Statistical Association, Sequential Analysis, Technometrics, Statistica Sinica, Statistics and Decisions, Journal of Statistical Planning and Inference, Metrika, Statistics and Probability Letters, Statistical Methodology, Communications in Statistics-Theory & Methods, Applied Stochastic Models, IEEE Transactions on Semiconductor Manufacturing, Stochastic Processes and their Applications, Statistics & Interface, Open Journal of Statistics, and Pakistan Journal of Statistics
- Reviewer of grant proposals for the National Science Foundation, the National Institutes of Health, and the Israeli Science Foundation (ISF)

University Service

- Graduate Education in Statistics working group, Committee Chair, 2017-2018
- Interdepartmental Data Science Working Group; Committee member, 2017-2018
- Member of Academic Senate, 1997-1998, 2001-2002
- Member of the IR committee, 1999-2001
- Chair of the IR committee, 2000-2001
- Member of the Wellness Committee, 2009-2011
- Member of the NS&M Faculty Peer Review Committee, 2009-2011
- Affiliate faculty of the American University Center for Innovation, 2016 - present

Departmental Service

At American University:

- Department Chair, 2022-2024
- Graduate Program Director in Statistics, 2016-2021, 2025 - Present
- Statistics Caucus, Chair, 2019-2021 (before 2019, informally)
- Graduate Studies Committee, Chair, 2016-2019
- Ph. D. program development, Committee Chair, 2019-2024
- Search Committee for Tenure-Line Faculty, 2017-18 (chair), 2018-19, 2025-26 (co-chair)
- Major Program Review Committee, Co-chair, 2020-2022
- Merit Review Committee, Committee Member, 2017-2019; Committee Chair, 2018-2019
- Departmental Honors coordinator, 2015-2019
- Mathematics and Statistics Colloquium, Co-Chair, 2017-2019
- Information Science track, Committee member, 2016-2017
- Data Science Track, Certificate, and Minor, Committee member, 2016-2017
- Rank and Tenure Committee, Committee member, 2014-present
- Developer of new courses (STAT 204, STAT 517 two topics courses, STAT 618, STAT 423/623, STAT 427/627, DATA 645 online)
- Mentoring colleagues, graduate students, and undergraduate students

At the University of Texas at Dallas:

- Graduate advisor in Statistics, 2004-2009

- MS and Fast Track advisor in Statistics, 2012-2014
- Undergraduate advisor in Statistics, 1997-2003
- MS and Fast Track advising committee (chair), 2012-2013
- Bylaws committee (chair), 2012-2014
- Teaching evaluation committee (chair), 2010-2012
- Web master for Math sciences, 1996-2006
- Graduate student admission and TA recruitment (chair), 1998-2001, 2004-2009
- Actuarial career coordinator, 2000-2002
- Undergraduate curriculum, 1997-2003, 2006-2014
- Graduate curriculum, 2004-2012
- Research mentoring and grant writing committee, 2012-2014 (chair, 2013 – 2014)
- Curriculum issues, 1998-1999
- Class schedule committee, 1999-2003, 2004-2009
- Planning committee, 1999-2001
- Academic seminars (co-chair), 1996-1997
- Statistical consulting service, 1995-1998
- Math sciences student computing lab, 1995-1996
- Computer facilities, 1998-2003, 2004-2014
- Service courses committee, 1998-2002
- Statistics service courses textbooks, 2002-2003, 2004-2006
- Statistics courses textbooks, 2002-2003, 2005-2014
- Library committee (chair), 2005-2006
- Ad Hoc committees (chair and member)
- T/TT and senior lecturer hiring committees (chair and member)
- Research mentor (2001 – 2014)

Personal

- U.S. Citizen
- Married. Children: Eric Abraham (1996), Anthony Nakhman (1999), Miriam Liba (2011), Natalia Dina (2014).
- Hobbies: I play piano, bridge, Go, and ice hockey; enjoy classical music and traveling